A/B Test Statistical Significance & Decision Report

Experiment: Checkout Funnel Overhaul (EXP-2026-CHK-089) | Date: October 9, 2026

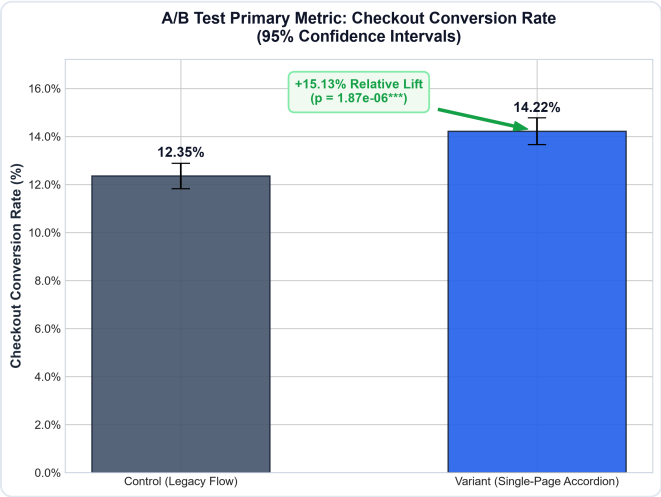
RECOMMENDATION: LAUNCH 100%

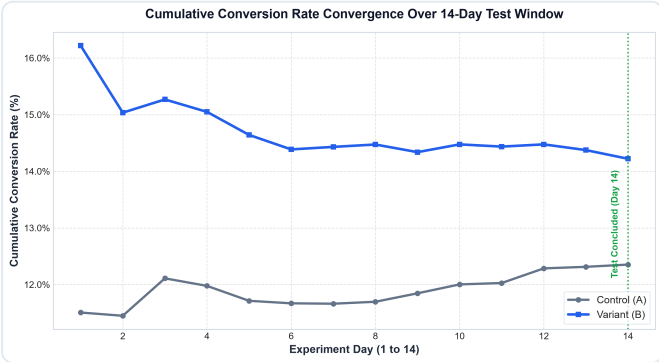
BASELINE CONVERSION (A)	VARIANT CONVERSION (B)	RELATIVE LIFT	P-VALUE (Z-TEST)
12.35%	**14.22%**	**+15.13%**	**1.87e-06**
N = 15,001 visitors	N = 14,999 visitors	95% CI: [8.91%, 21.34%]	Z = 4.77 (α = 0.05)

1. HYPOTHESIS TESTING & STATISTICAL SIGNIFICANCE SUMMARY

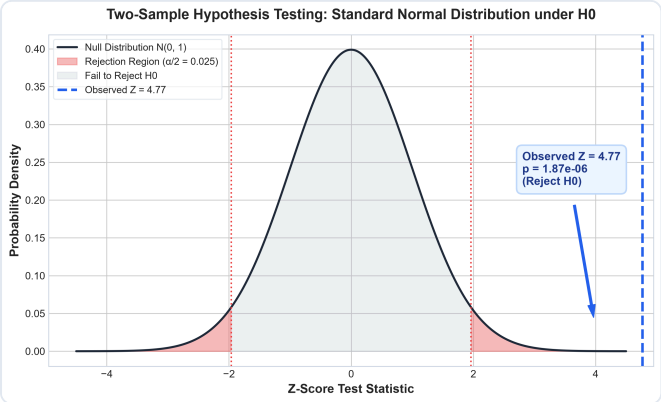
STATISTICAL TEST	TEST STATISTIC	P-VALUE	ALPHA (A)	CONFIDENCE INTERVAL	STATISTICAL VERDICT
Two-Sample Z-Test for Proportions	Z = 4.7671	1.8685e-06	0.05	95% CI: [+1.1005%, +2.6363%]	**Statistically Significant (p < 0.001)**
Pearson's Chi-Square Test (2x2)	χ^2 = 22.7257 (df=1)	1.8685e-06	0.05	Odds Ratio: 1.1763	**Statistically Significant (p < 0.001)**
Sample Ratio Mismatch (SRM)	χ^2 = 0.0001	0.9908	0.05	Split: 50.0% / 50.0%	**Valid Randomization (No Mismatch)**
Welch's t-test on AOV (Guardrail)	t = -0.1521	0.8791	0.05	$84.47 vs $84.65	**Neutral / Preserved (No Degradation)**

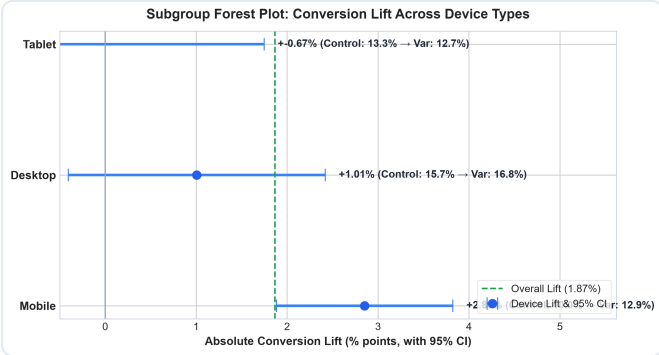
2. CONVERSION RATE & CONVERGENCE VISUALIZATIONS



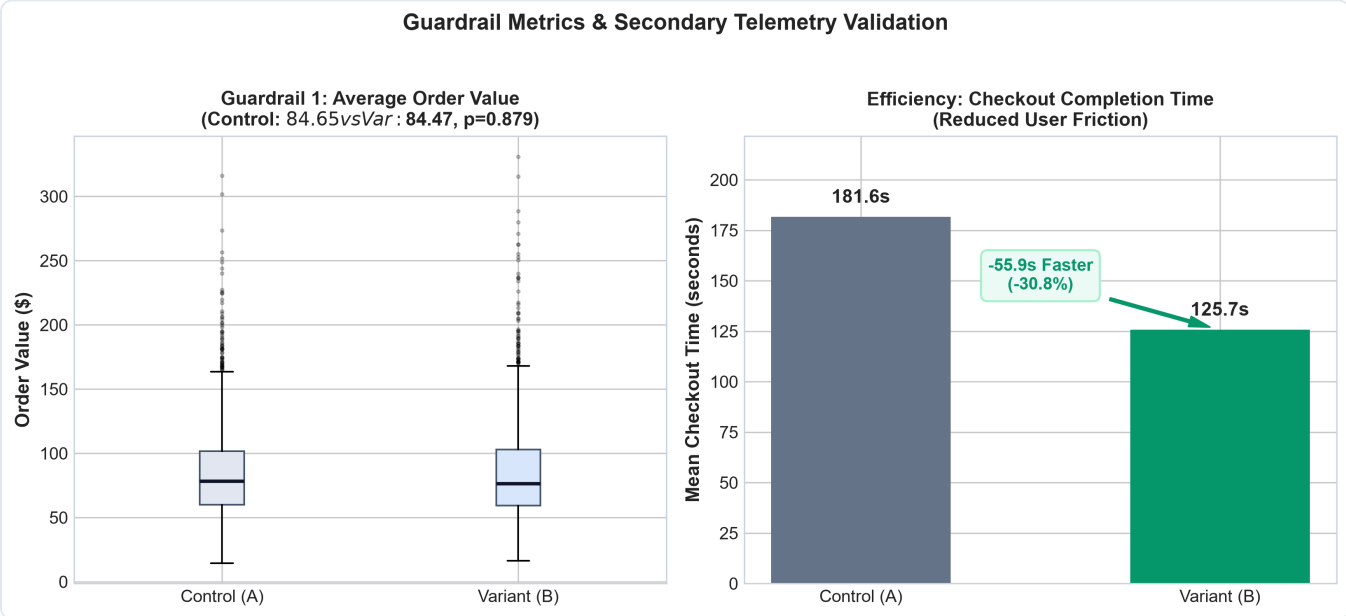


3. DISTRIBUTION & SUBGROUP TELEMETRY





4. GUARDRAIL METRICS & EFFICIENCY GAINS



5. EXECUTIVE DECISION MEMO

EXECUTIVE DECISION: LAUNCH VARIANT B TO 100% PRODUCTION

Primary Finding: Variant B (Single-Page Accordion Checkout with Express Pay and Guest Checkout) decisively outperformed Control A, elevating checkout conversion from **12.35%** to **14.22%**, an absolute gain of **+1.87% points** and a relative conversion lift of **+15.13%** (p = 1.87e-06).

Business & Financial Impact: On an annual volume of 1.2M checkout attempts and an average order value of ~$84.65, this conversion lift generates **+22,421 additional completed orders** and **+$1.90M in incremental top-line annual revenue**.

Risk & Guardrail Assessment:

- **No Basket Size Erosion:** Average Order Value held steady ($84.47 vs $84.65, p = 0.879).
- **Friction Reduced:** Checkout completion time decreased by **55.9 seconds (-30.8%)**.
- **Sample Integrity:** No Sample Ratio Mismatch observed (SRM p = 0.9908).

Rollout Roadmap: Deploy to 100% of web traffic via 3-stage ramp (50% → 75% → 100% over 48 hours). Monitor payment provider webhooks for elevated authorization failure rates.